Multi-Label Emotion Classification for Bantu Languages: A Comparative Study of TF-IDF and Multilingual Transformer Models

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Abstract

Emotion analysis in African languages remains a significantly under-explored area despite the continent's remarkable linguistic diversity. This paper investigates multi-label emotion classification across three typologically related Bantu languages—isiZulu, isiXhosa, and Swahili—using the BRIGHTER dataset [Abdulmumin et al., 2024]. We compare a TF-IDF with One-vs-Rest Logistic Regression baseline against two multilingual transformer models, mBERT [Devlin et al., 2019] and XLM-R [Conneau et al., 2020], for classifying six emotion categories (anger, disgust, fear, joy, sadness, surprise). The baseline achieves macro F1 of 0.25–0.27 across languages; somewhat unexpectedly, both transformer models fall below this on macro F1 across all three languages, suggesting that data volume rather than model complexity is the primary bottleneck in this setting. We document substantial class imbalance—particularly for disgust and fear—as a key challenge, and analyse cross-lingual differences in model behaviour. Our work contributes a reproducible multi-label emotion pipeline for low-resource Bantu languages and provides insights into how genealogical relatedness and data availability influence classifier performance.

1 Introduction

Emotion analysis is a branch of NLP concerned with automatically detecting affective states expressed in text. Applications span mental health monitoring, social media analysis, and opinion mining—areas with growing relevance for the African continent, where over 2 000 languages are spoken yet remain largely absent from mainstream NLP resources [Adelani et al., 2022].

Research has disproportionately focused on high-resource languages such as English and Mandarin, leaving African languages underserved in both data availability and model development. This resource gap is consequential: without effective tools for low-resource languages, NLP-based services remain inaccessible to hundreds of millions of speakers. Recent community efforts—including AfriSenti [Muhammad et al., 2023] and AfroXLMR [Alabi et al., 2022]—demonstrate growing momentum in African language NLP, yet multi-label emotion classification for Bantu languages remains largely unaddressed.

The BRIGHTER dataset [Abdulmumin et al., 2024] addresses this gap by providing multi-label emotion annotations for 28 languages, including several Bantu varieties. In this work, we restrict our experiments to three genealogically related Bantu languages—isiZulu (*zul*), isiXhosa (*xho*), and Swahili (*swa*)—to investigate whether intra-family linguistic relatedness influences the efficacy of multilingual pre-trained models.

Problem Statement. The core challenge we address is whether multilingual pre-trained models can classify emotions in Bantu languages when labelled training data is very limited and class imbalance is severe. These languages are morphologically complex and largely absent from transformer pre-training corpora, so it is not obvious that fine-tuning on a few hundred examples will outperform a simpler bag-of-words approach. Understanding when large models fail in genuinely low-resource settings is important for deciding how to invest in NLP tooling for underserved African language communities.

Research Questions.

1. How do mBERT and XLM-R compare to a TF-IDF baseline for multi-label emotion classification in Bantu languages, and does per-label class weighting improve minority-class recall?

*COS 760 Natural Language Processing — Department of Computer Science, University of Pretoria — S1 2026

2. How does performance vary across the three languages, and does the absence of an official training split for isiZulu and isiXhosa measurably harm performance relative to Swahili?
3. What prediction error patterns arise, and what do they reveal about model behaviour under class imbalance and data scarcity?

Contributions. We contribute: (1) a reproducible emotion classification pipeline for three Bantu languages; (2) a controlled TF-IDF vs. transformer comparison under matched conditions; and (3) analysis of class imbalance effects, error patterns, and cross-lingual variation in a low-resource setting. §2 surveys background; §3 details methodology; §4 presents experiments; §5 reflects on findings; §6 concludes.

Responsible NLP. This work addresses language underrepresentation by building pipelines for low-resource Bantu communities, contributing towards more equitable NLP.

2 Background

2.1 Multilingual Pre-trained Models

BERT [Devlin et al., 2019] introduced the transformer encoder architecture pre-trained with masked language modelling. Its multilingual variant, mBERT, extends this to 104 languages using a shared subword vocabulary, enabling zero-shot and few-shot cross-lingual transfer without language-specific fine-tuning. XLM-R [Conneau et al., 2020] improves on mBERT through a significantly larger and more diverse pre-training corpus (2.5 TB of filtered Common Crawl text across 100 languages), making it particularly effective for low-resource settings. Both models represent the current standard for cross-lingual NLP transfer. However, neither was pre-trained with explicit coverage of Bantu morphological structure, and their shared subword vocabularies may segment agglutinative forms in ways that obscure linguistically meaningful boundaries—a gap directly relevant to isiZulu and isiXhosa.

2.2 Emotion Analysis in African Languages

Sentiment and emotion analysis for African languages has grown substantially with the introduction of benchmarks such as AfriSenti [Muhammad et al., 2023], a multilingual Twitter sentiment dataset, and MasakhaNER [Adelani et al., 2022] for named entity recognition. Both use native-speaker annotations—important where cultural context shapes label interpretation—a standard BRIGHTER also follows. Mabokela et al. [2024] demonstrated that pre-trained language models substantially outperform classical approaches for low-resource African language sentiment, while Thangaraj et al. [2024] show that fine-tuned multilingual models achieve strong cross-lingual transfer between related low-resource Bantu languages. However, these studies focus on single-label sentiment or sequence-level tasks; multi-label emotion classification under severe data constraints (fewer than 900 training examples) has received little systematic attention for Bantu languages, representing the gap this work addresses.

2.3 Multi-label Emotion Classification

Unlike single-label sentiment, multi-label emotion classification assigns a set of emotion labels to each text instance, reflecting the co-occurrence of emotions in natural language. Standard approaches include binary relevance (one classifier per label), label-powerset methods, and sigmoid-activated transformer classifiers with binary cross-entropy loss [Devlin et al., 2019, Conneau et al., 2020], though most benchmarks are English-centric with little evaluation under the data scarcity characteristic of low-resource African languages.

2.4 The BRIGHTER Dataset

BRIGHTER [Abdulmumin et al., 2024] provides multi-label emotion annotations across 32 languages, with annotations sourced from fluent speakers. The dataset includes six emotion categories—anger, disgust, fear, joy, sadness, and surprise—encoded as binary label vectors, making it directly applicable to multi-label classification.

Gap. Prior work has focused on broad cross-continental transfer without exploiting intra-family linguistic relatedness. This project addresses that gap by controlling language selection to a single genealogical family (Bantu) and examining how data availability interacts with model performance in a low-resource setting.

Responsible NLP. Research priorities in African language NLP are often shaped by data availability rather than community need; pre-training corpora for mBERT and XLM-R heavily favour high-resource European languages, which may limit their coverage of Bantu morphology—a constraint we acknowledge throughout.

3 Methodology

3.1 Data

We use the BRIGHTER dataset for three Bantu languages. Table 1 summarises split sizes and the emotion label distribution in the training split for each language. isiZulu and isiXhosa lack an official training partition; following standard practice in low-resource NLP, we use the development (validation) split as training data and evaluate on the held-out test set.

Lang	Train	Dev	Test	Train src.
isiZulu	—	875	2,047	validation
isiXhosa	—	682	1,594	validation
Swahili	3,307	1,102	3,312	train

Table 1: Dataset split sizes. isiZulu and isiXhosa have no official train split; the validation set is used as training data.

Label distribution is highly skewed. For isiXhosa, *sadness* accounts for 41.2% of training labels while *disgust* appears in only 0.4% (3 samples)—a 93.7:1 imbalance ratio. isiZulu shows moderate imbalance (7.2:1), and Swahili's most severe case is *fear* at 2.8% of labels (93 samples, ratio 5.8:1).

Exploratory data analysis. Swahili text is predominantly ASCII-compatible; isiZulu and isiXhosa encode click consonants via digraphs (*hl*, *xh*, *gc*) that mBERT and XLM-R tokenisers frequently split across sub-word boundaries, losing morphologically significant information. Minor script inconsistencies in isiXhosa (e.g. inconsistent apostrophe usage for clicks) reflect dialectal variation in the social-media source text.

Responsible NLP. BRIGHTER is publicly released; no proprietary or personal data is used. Using the development split as training data blurs the train–validation boundary and likely underestimates achievable results.

3.2 Preprocessing

Text is used without stemming or lemmatisation, preserving morphological richness important in Bantu languages. For the TF-IDF baseline, accents are retained (`strip_accents=None`) to avoid losing language-specific characters. Transformer models apply their native subword tokenisation (WordPiece for mBERT; SentencePiece for XLM-R) with maximum sequence length 128.

3.3 Models

TF-IDF Baseline. A TF-IDF vectoriser (maximum 5,000 features, unigrams and bigrams, minimum document frequency 2) feeds a One-vs-Rest Logistic Regression classifier (`class_weight="balanced"`, `max_iter=1000`) [Pedregosa et al., 2011]. The `balanced` weighting compensates for label imbalance by inversely weighting class frequencies.

mBERT. `bert-base-multilingual-cased` [Devlin et al., 2019] is fine-tuned with a multi-label classification head. The `problem_type="multi_label_classification"` configuration applies binary cross-entropy with logits (BCEWithLogitsLoss) natively.

XLM-R. `xlm-roberta-base` [Conneau et al., 2020] uses an identical head and loss configuration to mBERT, enabling direct comparison.

3.4 Training and Evaluation Setup

All transformer models are fine-tuned using the HuggingFace Transformers library [Wolf et al., 2020] under identical, controlled conditions to ensure fair comparison: 3 epochs, learning rate 2×10^{-5}, batch size 8, weight decay 0.01, seed 42. At inference, logits are converted to probabilities via sigmoid and binarised at threshold $\tau = 0.3$—a value lower than the standard 0.5 to improve recall on infrequent emotion classes.

Evaluation metrics. Macro F1 is the primary metric, as it treats all classes equally and is robust to class imbalance. Supplementary metrics include micro F1, macro precision, macro recall, Hamming loss, and macro Jaccard score.

Error analysis methodology. To directly address RQ3, we manually inspect test-set predictions and categorise errors as: false negatives (model predicts nothing despite a positive ground-truth label), over-predictions (model predicts a label absent from the ground truth), or partial matches (some labels correct, others missed or spurious). We additionally flag any per-class F1 below 0.15 as a *weak label*, indicating a systematic failure to learn that emotion class.

4 Experiments and Results

4.1 Baseline Performance

Table 2 reports TF-IDF + Logistic Regression results. Macro F1 scores range from 0.250 (Swahili) to 0.265 (isiXhosa). Despite isiXhosa and isiZulu using the development set as training data (limiting training size to 682 and 875 samples respectively), their macro F1 is comparable to Swahili, which has 3,307 training samples. However, Swahili achieves a notably lower micro F1 (0.297), suggesting its predictions are more uniformly spread but less precise on frequent classes.

Language	F1-mac	F1-mic	Prec-mac	Rec-mac
isiZulu	0.257	0.341	0.235	0.290
isiXhosa	0.265	0.471	0.247	0.321
Swahili	0.250	0.297	0.215	0.304

Table 2: TF-IDF + Logistic Regression baseline results.

4.2 Transformer Results

Tables 3 and 4 report fine-tuned mBERT and XLM-R performance.

The mBERT results, fine-tuned with per-label positive-class weighting (`pos_weight`), show improved recall on minority classes compared to the unweighted configuration. isiXhosa (682 training samples) achieves macro F1 of 0.240, broadly comparable to its TF-IDF baseline (0.265), while recall improves markedly from 0.321 to 0.530 — indicating that class weighting successfully pushes predictions towards minority labels. isiZulu (875 training samples) achieves macro F1 of 0.177, below the TF-IDF baseline (0.257), but recall rises from 0.290 to 0.560; the remaining F1 gap reflects severe sparsity of disgust (2.9%) and fear (3.1%) in isiZulu training. Swahili, with the largest training set (3,307 examples), achieves macro F1 of 0.193 — still below its TF-IDF baseline of 0.250, but with recall of 0.769, reflecting the weighted loss effectively promoting minority-class predictions at the cost of precision (0.112).

Language	F1-mac	F1-mic	Prec-mac	Rec-mac
isiZulu	0.177	0.259	0.124	0.560
isiXhosa	0.240	0.439	0.159	0.530
Swahili	0.193	0.232	0.112	0.769

Table 3: mBERT fine-tuning results (3 epochs, batch 8, τ=0.3).

Language	F1-mac	F1-mic	Prec-mac	Rec-mac
isiZulu	0.135	0.218	0.079	0.547
isiXhosa	0.222	0.447	0.147	0.500
Swahili	0.223	0.249	0.136	0.765

Table 4: XLM-R fine-tuning results (3 epochs, batch 8, τ=0.3).

XLM-R, also fine-tuned with per-label class weighting, follows a broadly similar pattern to mBERT. For isiZulu (875 samples), class weighting prevents collapse, yielding F1 0.135 and recall 0.547—below mBERT (0.177) due to XLM-R's larger parameter count (278M vs. 178M); isiXhosa (682 samples) scores F1 0.222, slightly below mBERT's 0.240. For Swahili, XLM-R (F1 = 0.223) narrows the gap with mBERT (F1 = 0.193) when trained on 3,307 examples, confirming that XLM-R benefits more from additional training data. Notably, the TF-IDF baseline (F1 = 0.250–0.265) outperforms both transformer models across all three languages under these data constraints, highlighting the challenge of deploying large pretrained transformers in genuinely low-resource settings without dedicated hyperparameter tuning per language. This appears to contradict Mabokela et al. [2024]'s finding that pre-trained models outperform classical methods, but those experiments use thousands of training examples; our sub-900-sample constraint eliminates the transformer advantage, consistent with Thangaraj et al. [2024]'s finding that multilingual models require fine-tuning to realise cross-lingual gains in low-resource settings.

4.3 Weak Label Analysis

Table 5 shows the five emotion labels identified as weak (macro F1 < 0.15) in the baseline model. All weak labels correspond to the rarest classes in their respective language splits.

Lang	Label	F1	Recall	Support
isiXhosa	disgust	0.017	0.125	8
isiXhosa	fear	0.044	0.065	31
isiXhosa	anger	0.069	0.091	66
isiZulu	disgust	0.075	0.119	59
Swahili	fear	0.069	0.064	94

Table 5: Weak labels (F1 < 0.15) from the TF-IDF baseline. Support reflects test-set samples.

isiXhosa *disgust* is the most extreme case: despite 8 test samples, the model achieves only F1 = 0.017, with the extremely limited training exposure (3 training samples) making generalisation essentially impossible.

4.4 Error Analysis

We categorised prediction errors into three types based on the test set:

- **False negatives**: the model predicts no labels despite ground truth labels being present. This is the most common failure mode across all three languages.
- **Over-prediction**: the model predicts labels that do not appear in the ground truth.
- **Partial match**: some labels are predicted correctly but additional labels are missed or spuriously added.

False negatives dominate across isiZulu, isiXhosa, and Swahili. A representative example from isiZulu illustrates the challenge: the text *"bayaveva ngokujola kukathisha nomfundi"* (expressing anger about a teacher–student relationship) receives no prediction, likely because TF-IDF features cannot generalise the infrequent vocabulary. Over-prediction errors tend to co-occur with sadness and disgust, suggesting the model associates surface-level negative words with multiple emotion classes simultaneously.

Responsible NLP. isiXhosa achieves higher micro F1 (0.471) than isiZulu (0.341) despite fewer training samples, driven by its skewed label distribution (sadness + joy at 72.3%)—illustrating how aggregate metrics obscure per-label disparities and that evaluation must be disaggregated.

Answers to research questions. (RQ1) TF-IDF outperforms both transformers on macro F1, showing large models are not automatically superior under data scarcity. (RQ2) Performance tracks data volume rather than linguistic distance: Swahili's larger corpus yields better precision–recall tradeoffs; isiZulu and isiXhosa suffer most from lacking official training splits. (RQ3) False negatives dominate, driven by class sparsity on rare emotions.

5 Reflections and Discussion

5.1 What Worked

The TF-IDF + Logistic Regression baseline provides a consistent, reproducible lower bound and is computationally accessible on consumer hardware. The `class_weight="balanced"` configuration mitigated—though did not eliminate—the effect of class imbalance. The modular pipeline ensures baseline and transformer experiments share data loading and evaluation utilities.

5.2 Challenges

Limited training data. isiZulu and isiXhosa lack official training splits in BRIGHTER, forcing use of the development set as a training surrogate. With only 682–875 samples, models are severely data-constrained for a six-class multi-label problem.

Class imbalance. isiXhosa *disgust* appears in 0.4% of training samples (3 instances), making it statistically impossible for any model to learn a meaningful decision boundary. Even Swahili—with the largest training set—has *fear* at only 2.8%.

Computational constraints. Transformer fine-tuning was conducted on CPU, severely limiting feasible configurations; initial experiments with one epoch and batch size 2 collapsed to all-zero predictions until increasing to 3 epochs, batch size 8, and threshold 0.3. Concrete extensions addressing these limitations are discussed in Section 6.

5.3 Responsible AI Principles

Data openness. BRIGHTER and all project code are publicly available and version-controlled.

Fairness across languages. The controlled selection of Bantu languages was an intentional choice to study cross-lingual fairness within a genealogical family. Results show that performance inequality exists even among related languages, driven primarily by training data volume and class distribution rather than linguistic distance alone.

Limitations and risks. The models evaluated here are research prototypes not suitable for deployment. Emotion classification carries inherent risks of misclassification, which could have adverse consequences in sensitive applications such as mental health monitoring. The near-zero performance on rare emotions (disgust, fear) means these classes are effectively undetected by all models evaluated.

Broader societal impact. Building emotion analysis infrastructure for Bantu languages has positive long-term implications for social media analysis, public health surveillance, and community voice amplification. However, any deployment should involve community consultation and ongoing fairness auditing.

6 Conclusion

We presented a comparative evaluation of TF-IDF and multilingual transformer models for multi-label emotion classification in three Bantu languages (isiZulu, isiXhosa, Swahili) using the BRIGHTER dataset.

The TF-IDF + Logistic Regression baseline achieves macro F1 of 0.250–0.265, establishing a reproducible lower bound. Key findings include: (1) severe class imbalance—particularly for *disgust* and *fear*—substantially harms recall on minority classes; (2) false negatives are the dominant error type, suggesting models default to abstaining rather than over-predicting; (3) the TF-IDF baseline outperforms both transformer models on macro F1 across all three languages under the data constraints of this study.

For the transformer experiments, fine-tuned with per-label positive-class weighting (`pos_weight`) to address minority-class collapse, mBERT and XLM-R exhibit complementary data-sensitivity profiles. mBERT (178M parameters) achieves macro F1 of 0.177 (isiZulu) and 0.240 (isiXhosa), while XLM-R (278M parameters) scores 0.135 and 0.222 respectively, confirming that the larger model is more data-hungry in low-resource conditions. Both models substantially improve recall over the baseline (mBERT recall: 0.530–0.769 vs. baseline 0.290–0.321), but at the cost of precision, reflecting the `pos_weight` trade-off between sensitivity and specificity.

These findings underscore that data volume is the primary bottleneck for emotion analysis in low-resource Bantu languages, and that larger pretrained models are not automatically superior when training data is scarce. Future work should explore data augmentation via back-translation or cross-lingual projection from Swahili to isiZulu and isiXhosa, fine-tuning AfroXLMR [Alabi et al., 2022] which is better adapted for Bantu morphology, and per-class threshold tuning to improve the precision–recall balance on rare emotions.

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